

Lab 6: ARCH effects, GARCH and Value-at-Risk

Reference solution

Aymen Ben Brik
aymen.benbrik@esprit.tn
Esprit School of Business

2025–2026

Reference Python code in `correction.py`. The synthetic dataset is regenerated by `generate_returns.py` (GARCH(1, 1) with $(\alpha_0, \alpha_1, \beta_1) = (0.05, 0.10, 0.85)$, $T = 1500$, seed 42).

Exercise 1 – Volatility clustering

Solution

$T = 1500$, sample mean = +0.0454, sample sd = 0.9515, excess kurtosis = 0.44. The time plot shows quiet stretches alternating with bursts. The ACF of r_t stays inside the Bartlett band, but the ACF of r_t^2 has many large positive spikes – the *volatility clustering* signature. Returns are serially uncorrelated but *not* independent.

Exercise 2 – Engle’s LM test

Solution

lag q	LM stat	$\chi^2_{q,0.95}$	p -value	Decision
5	158.57	11.07	≈ 0	reject – ARCH
10	162.86	18.31	≈ 0	reject – ARCH
20	172.66	31.41	≈ 0	reject – ARCH

$LM \gg \chi^2_{q,0.95}$ at every lag: strong evidence of ARCH effects, consistent with the heavy ACF spikes on r_t^2 .

Exercise 3 – ARCH(q) fitting

Solution

q	log-lik.	AIC	$\sum_i \hat{\alpha}_i$	First $\hat{\alpha}_i$ ’s
1	−2033.84	4073.68	0.159	0.159
3	−2020.11	4050.23	0.299	0.105, 0.093, 0.101
5	−2007.27	4028.54	0.448	0.078, 0.081, 0.085, 0.107, 0.096

$\sum \hat{\alpha}_i < 1$ in every case (covariance-stationary). ARCH(5) improves over ARCH(3) but the gain is modest: the spread of the loadings shows that pure ARCH cannot capture the *long memory* of the conditional variance with few parameters.

Exercise 4 – GARCH(1, 1) fitting**Solution**

MLE estimates with constant mean and Gaussian innovations:

$$\hat{\mu} = +0.0488, \quad \hat{\alpha}_0 = 0.0878, \quad \hat{\alpha}_1 = 0.1069, \quad \hat{\beta}_1 = 0.7960.$$

True DGP: (0.05, 0.10, 0.85) – estimates are within sampling error. Persistence:

$$\hat{\alpha}_1 + \hat{\beta}_1 = 0.9029 < 1 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{0.0878}{1 - 0.9029} = 0.9045,$$

to be compared with the sample variance 0.9053 – excellent match.

GARCH(1, 1): $\log \mathcal{L} = -2009.31$, AIC = 4026.62. ARCH(5) has AIC = 4028.54. **GARCH(1, 1) wins on AIC with three parameters (instead of six):** parsimony argument confirmed.

Exercise 5 – Standardised-residual diagnostics**Solution**

On $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$:

lag q	LM stat (on $\hat{z}_t^2$)	p -value	Decision
5	4.67	0.457	do not reject – white
10	9.01	0.531	do not reject – white
20	15.61	0.740	do not reject – white

The Engle LM no longer rejects at any lag: GARCH(1, 1) has absorbed the volatility clustering. The QQ-plot of $\hat{z}_t$ versus $\mathcal{N}(0, 1)$ is approximately linear, consistent with the simulated Gaussian innovations.

Exercise 6 – Forecast and Value-at-Risk**Solution**

The closed-form mean-reversion identity

$$\hat{\sigma}_{T+h}^2 = \hat{\sigma}^2 + (\hat{\alpha}_1 + \hat{\beta}_1)^{h-1} (\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$$

matches the `arch` package output to machine precision (max discrepancy 6.7×10^{-16}). The figure `lab6_ex6_forecast.png` shows the slow decay toward $\hat{\sigma}^2 = 0.9045$ at rate 0.9029.

Value-at-Risk. At T , the conditional sd is $\hat{\sigma}_{T+1} = 1.339$, larger than the unconditional sd 0.951: the series ends in a turbulent regime. Resulting one-day VaR:

α	z_α	VaR conditional	VaR unconditional
1%	-2.326	3.07	2.16
5%	-1.645	2.15	1.52

The conditional VaR is *larger* – correctly flagging higher short-term risk that the static VaR misses.